

Fragmentation

Category 1

Task

The disk (HDD, SSD) in your laptop is divided into clusters. Typically, a cluster is 4096 bytes in size.

When you write a file to your disk, the operating system looks for enough unused clusters to store the entire file. For example, suppose the file is $5 \times 4096 = 20480$ bytes in size, then it fits in 5 clusters.

But what if the file size is not a multiple of 4096? Suppose it is 20481 bytes? Clearly, 5 clusters are no longer sufficient. Can a file occupy just part of a cluster? For efficiency reasons, this is not allowed: each file must occupy a whole number of clusters. Clusters also cannot be shared by multiple files. This results in some disk space being wasted: a file of $5 \times 4096 + 1 = 20481$ bytes will take 6 clusters, which equals $6 \times 4096 = 24576$ bytes. The file thus uses 24576 bytes of disk space, even though it is only 20481 bytes in size. This loss is called internal fragmentation.

For this task, you must calculate, given the sizes of a number of files, how much disk space they occupy, taking internal fragmentation into account.

Input

The first line contains a positive integer giving the number of test cases. Then, for each test case, there is one line with a sequence of positive integers separated by a space. The first number indicates how many numbers follow on that same line. The remaining numbers represent the file sizes.

Example Input

```
6
1 20480
1 20481
1 1
1 0
1 861685
3 1 1 1
```

Output

For each test case, output one line consisting of two numbers separated by a space. The first number is the index of the test case, starting at 1. The second number is how much disk space is required to store all the files.

Example Output

1 20480

2 24576

3 4096

4 0

5 864256

6 12288